

Prisha Bhatia

prishabhatia4444@gmail.com | +91 8198002511 | [GitHub](#) | [LinkedIn](#) | [LeetCode](#)

EDUCATION

Vellore Institute of Technology, Bhopal <i>B.Tech in Computer Science (AI & ML)</i>	Sep 2023 – Present <i>CGPA: 9.13 / 10</i>
Anand Public Sr Sec School, Kapurthala <i>Class XII</i>	2023 89%
Christ King Convent School, Kapurthala <i>Class X</i>	2021 94%

TECHNICAL SKILLS

Languages: Python, C++, Java
Python Ecosystem: NumPy, Pandas, NetworkX, PyVis, Scikit-learn, PyTorch Geometric, Flask, Tkinter, Subprocess
Core CS: Data Structures and Algorithms, OOP, Graph Theory, Clustering Algorithms
Systems and Tools: Linux, WSL2, Git, MySQL, REST APIs, AWS (EC2, S3, IAM)
Web: HTML, CSS, Jinja2

EXPERIENCE

FOSSEE, IIT Bombay <i>Python Developer, Semester Long Intern, eSim KiCad Plugin (Remote/Hybrid)</i>	Feb 2026 – May 2026
- Built PCB Linter, a 1500+ line Python tool automating PCB circuit analysis using graph algorithms and clustering; processes KiCad board files via Python scripting API - Implemented Deterministic Annealing clustering from scratch in Python; auto-groups circuit nodes into voltage, current, and power domains using similarity matrices and temperature-based annealing - Engineered a SPICE netlist generator in Python that parses PCB component data, sanitises net names, maps 10+ component types, and auto-injects subcircuit models from a 647-file library - Built a universal file parser in Python handling 4 simulation output formats (DC, AC, Transient, CSV) with automatic format detection using regex and string processing - Used NetworkX for PCB connectivity graph construction; computed degree centrality and betweenness centrality to detect high-fanout nodes and single points of failure - Generated dynamic HTML reports via Python string templating with interactive clustering tabs, heatmaps, voltage attenuation matrices, and dark/light mode toggle - Automated end-to-end pipeline: PCB parsing, SPICE generation, NGSpice subprocess execution, output parsing, clustering, and report generation, entirely in Python	
FOSSEE, IIT Bombay <i>Python Developer, eSim Summer Fellow, KiCad Plugin (Remote/Hybrid)</i>	May 2026 – Present

PROJECTS

eSim Tool Manager <i>Python, subprocess, apt, winget</i>	Jan 2026 – Feb 2026
- Built a cross-platform Python CLI tool for automated installation and configuration of eSim and all dependencies on Linux and Windows - Integrated apt and winget package managers via Python subprocess calls; implemented version detection, PATH configuration, and automated upgrade workflows - Reduced manual setup time significantly; adopted by FOSSEE team for intern onboarding	
Power Consumption Analysis <i>Python, Flask, Scikit-learn, Matplotlib</i>	May 2025 – Jul 2025
- Built an ML-powered web dashboard using Flask to analyze and visualize household energy consumption patterns - Performed data preprocessing, feature engineering, and trained regression models to identify consumption inefficiencies - Deployed interactive charts using Matplotlib and Jinja2 templates for real-time data exploration	
Crop Recommendation System <i>Python, Scikit-learn, Pandas</i>	Feb 2025 – Apr 2025
- Developed ML model for crop recommendation based on soil and climate data; improved accuracy using feature engineering and cross-validation - Built data pipeline in Python for cleaning, encoding, and transforming agricultural datasets for model training	

ACHIEVEMENTS & CERTIFICATIONS

Machine Learning Specialization, University of Michigan (Coursera) | Microsoft Azure Data Fundamentals | IBM SkillsBuild AI Certification | AWS Cloud Practitioner and Solutions Architect | Solved 100+ DSA problems on LeetCode

LEADERSHIP

Cricket Team Captain, VIT Bhopal | Participant, All India ICSE Frank Anthony Debate Competition | Head Girl, Christ King Convent School